



CANDIDATE  
NAME

**ANSWER KEY**

CIVICS  
GROUP

2

0

-

REGISTRATION  
NUMBER

**BIOLOGY**

**9744/04**

Paper 4 Practical

**24 August 2021**

**2 hours 30 minutes**

Candidates answer on the Question Paper.

**READ THESE INSTRUCTIONS FIRST**

Write your name, civics group and registration number on all the work you hand in.  
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.  
Write in dark blue or black pen.  
You may use an HB pencil for any diagrams or graphs.  
Do not use stapler, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided in the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.  
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

|                   |
|-------------------|
| <b>Shift</b>      |
|                   |
| <b>Laboratory</b> |
|                   |

| For Examiner's Use |    |
|--------------------|----|
| 1                  |    |
| 2                  |    |
| Total              | 55 |

This document consists of **16** printed pages.

Answer **all** questions.

Plan the use of the time given to ensure that you finish the whole of Question 1 and Question 2.

- 1** Yeast cells contain respiratory enzymes that catalyses the oxidation of sugars to release energy. During these reactions, carbon dioxide is released. Some of this carbon dioxide dissolves in water forming carbonic acid.

The progress of these enzyme-catalysed reactions can be followed by measuring the time taken for an indicator to change colour as carbonic acid is formed. This gives an indication of the rate of respiration in yeast cells.

You will investigate the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

You are provided with the materials shown in Table 1.1.

**Table 1.1**

| labelled | contents                             | hazard       | volume/cm <sup>3</sup> |
|----------|--------------------------------------|--------------|------------------------|
| <b>Y</b> | 10% yeast suspension                 | none         | 100                    |
| <b>H</b> | hydrogencarbonate indicator solution | none         | 50                     |
| <b>C</b> | carbonic acid solution               | eye irritant | 5                      |
| <b>W</b> | distilled water                      | none         | 100                    |

You are also provided with:

- a beaker, labelled **hot water**, in which to collect hot water at temperature of 50 °C to 55 °C
- a beaker, labelled **tap water**, in which to collect tap water

If any solution comes into contact with your skin, wash it off immediately under cold water. It is recommended that you wear suitable eye protection. Care should be taken when using hot water.

The hydrogencarbonate indicator solution, **H**, will change colour from red to yellow as carbon dioxide passes through it.

To help you identify the end-point, carry out step 1 to step 3.

- 1** Put 2 cm<sup>3</sup> of **H** into a small test-tube.
- 2** Add 1 drop of **C** into this test-tube and mix.
- 3** The colour of the indicator solution will change from red to yellow. This is the end-point.

You will need to use this test-tube for colour comparison.

- (a) You will use **proportional dilution** to make different concentrations of yeast suspension, **Y**.

You will need to prepare 25 cm<sup>3</sup> of each concentration of yeast suspension.

- (i) State the concentrations of yeast suspension that you will use in your investigation. [1]

**Five concentrations at equal intervals:** 2.0%, 4.0%, 6.0%, 8.0%, 10.0%

**R:** serial dilution

**I:** 0.0%

- (ii) Complete Table 1.2 to show how you will prepare the concentrations of yeast suspension you have listed in **(a)(i)**. [2]

**Table 1.2**

| concentration of yeast suspension /% | volume of <b>Y</b> /cm <sup>3</sup> | volume of <b>W</b> /cm <sup>3</sup> |
|--------------------------------------|-------------------------------------|-------------------------------------|
| 10.0                                 | 25.0                                | 0.0                                 |
| 8.0                                  | 20.0                                | 5.0                                 |
| 6.0                                  | 15.0                                | 10.0                                |
| 4.0                                  | 10.0                                | 15.0                                |
| 2.0                                  | 5.0                                 | 20.0                                |

- Correct values to make up concentrations in first column;
- Volume to 1 decimal place; **R:** units in table

- 4 Use a glass rod to stir the yeast suspension, **Y**, to suspend the yeast cells.  
When using a syringe to transfer **Y**, remove **Y** from below any froth on the surface.
- 5 Prepare the concentrations of yeast suspension as stated in **(a)(i)** and shown in Table 1.2.

A student carried out some preliminary experiments to find the optimum conditions for investigating the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

The student found that:

- the yeast suspension should **not** be mixed directly with hydrogencarbonate indicator, as the colour change was difficult to observe
- the optimal temperature for yeast enzyme activity is 40 °C
- the best volume of yeast suspension to use is 20 cm<sup>3</sup>
- the best volume of hydrogencarbonate indicator to use is 2 cm<sup>3</sup>.

- (iii) Use the information provided to plan a method that you can use to investigate the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour. Do **not** plan to carry out repeats. You will carry out this method in step 6.

In your plan you must use:

- the different concentrations of yeast suspension from step 5
- the test-tube showing the end-point colour from step 3
- hydrogencarbonate indicator solution.

You may select from any of the other **apparatus and materials provided**.

**Note:** this means you are **NOT** allowed to use other apparatus in your planned method!

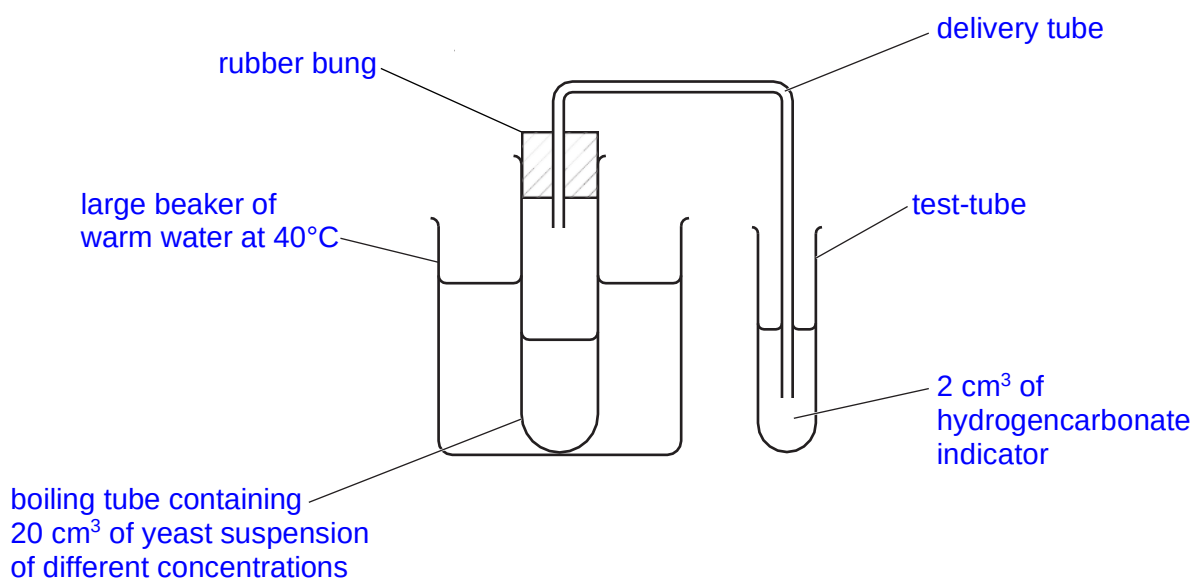
Your planned method should:

- have a clear and helpful structure so that the method described could be repeated by anyone reading it
- include helpful details to ensure that results are as accurate and repeatable as possible
- use the correct technical and scientific terms
- only make use of the apparatus and materials provided.

[6]

Marking points:

1. **Experiment setup:** yeast suspension in boiling tube and connected to test-tube containing hydrogencarbonate indicator via delivery tube (give credit for appropriately labelled diagram);  
R: adding hydrogencarbonate directly into yeast suspension;
2. Correct volumes of yeast suspension ( $20\text{ cm}^3$ ) and hydrogencarbonate indicator ( $2\text{ cm}^3$ ) used + measured using appropriate instruments, i.e.  $10\text{ cm}^3$  and  $1\text{ cm}^3$  syringe respectively or  $10\text{ cm}^3$  syringe to measure both volumes;  
R: diluting hydrogencarbonate with distilled water  
R: mention of  $5\text{ cm}^3$ ,  $12\text{ cm}^3$  syringe
3. Setup of water-bath at  $40^\circ\text{C}$  + description of how to maintain temperature by adding hot water and tap water as necessary, and monitor using thermometer;
4. **Equilibration time** of yeast suspension in water bath for 2 minutes – to allow yeast suspension to reach same temperature as the water bath; (A: at least 1 minute)
5. Measure time taken for hydrogencarbonate indicator to change from red to yellow, using test-tube showing the end-point colour (from step 3) for colour comparison;
6. Repeat steps for all other concentrations of yeast suspension;



Before proceeding further, use the beaker labelled **hot water** to collect approximately 400 cm<sup>3</sup> of hot water from where it is provided in the laboratory.

The beaker labelled **tap water** can be filled as necessary from the tap.

**Proceed as follows.**

- 6 Carry out your method described in **(a)(iii)** to investigate the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour.

Record your results in **(a)(iv)**.

If no colour change occurs with hydrogencarbonate indicator solution after three minutes, record the result as 'more than 180'.

- (iv)** Use the space provided to record your results from step 6 in a suitable format. [3]

| concentration of yeast suspension / % | time taken for hydrogencarbonate indicator to turn from red to yellow / s<br>R: time / s |
|---------------------------------------|------------------------------------------------------------------------------------------|
| 10.0                                  | 48                                                                                       |
| 8.0                                   | 115                                                                                      |
| 6.0                                   | 170                                                                                      |
| 4.0                                   | more than 180                                                                            |
| 2.0                                   | more than 180                                                                            |

- Results presented in table form with appropriate headings + units; R: units in table
- All concentrations have data recorded + time taken recorded in seconds with precision as whole number + time taken that is more than 3 minutes must be recorded as "more than 180";
- Value recorded for at least 10.0% yeast suspension must be lower than 8.0%;  
R:  
  - inaccurate readings such as single digit for time taken, e.g. 2s for 10.0% yeast suspension
  - If 2.0% yeast suspension is not recorded as "more than 180"

- (v)** From your results, state a valid conclusion about the effect of changing the concentration of yeast suspension on the time taken for the indicator to change colour. [1]

The higher the concentration of yeast suspension, the less/shorter time taken for the indicator to change colour from red to yellow (or vice versa);

- (vi)** Identify one significant source of error in this investigation. Give a reason for your answer. [2]

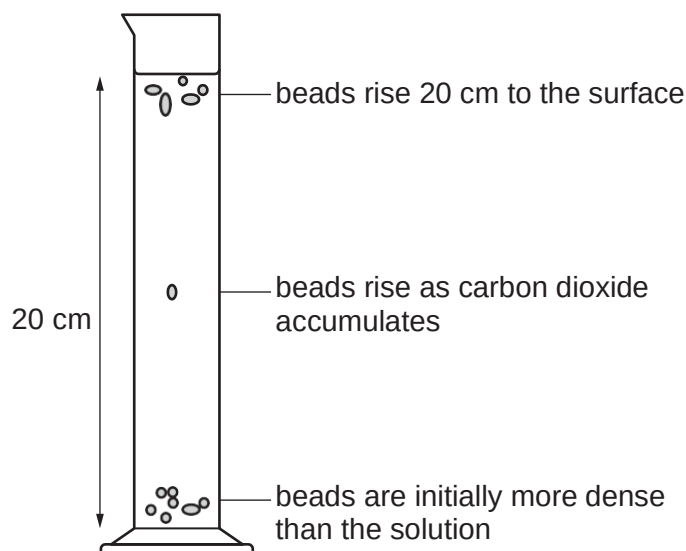
| source of error                                                                                                                            | reason                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| difficulty in determining end point colour<br>A: determination of end point is inaccurate<br>R: time taken for colour change is inaccurate | <u>colour identification</u> using naked eye is <u>subjective</u> ; (can give ECF even if source of error phrased wrongly) |
| temperature of water bath fluctuates / difficult to maintain                                                                               | Time taken for colour change will be inaccurate / rate of reaction is affected                                             |

A: substrate concentration not constant for all yeast concentrations

- (b) A student carried out an investigation into the metabolic activity of enzymes in 5 different species of yeast, **P** to **T**.

The student immobilised the yeast cells in alginate beads, which were then dropped into a solution of the substrate at the same temperature.

As carbon dioxide collected in the beads, they rose to the surface of the solution of the substrate as shown in Fig. 1.1.



**Fig. 1.1**

The student measured the time taken for 5 beads to rise to the surface, a distance of 20 cm.

The processed results for each species are shown in Table 1.3.

**Table 1.3**

| species  | mean time to rise/s |
|----------|---------------------|
| <b>P</b> | 72                  |
| <b>Q</b> | 19                  |
| <b>R</b> | 11                  |
| <b>S</b> | 36                  |
| <b>T</b> | 108                 |

- (i) Identify the species with the most active metabolic enzymes ..... **R** ..... [1]

For this species, calculate the rate at which these beads rise 20 cm.

1. Rate =  $1 / \text{mean time to rise}$   
 $= 1 / 11 \text{ s}$   
 $= 0.0909 \text{ s}^{-1}$

OR 1. Rate =  $20 \text{ cm} / \text{mean time to rise}$   
 $= 20 \text{ cm} / 11 \text{ s}$   
 $= 1.82 \text{ cm s}^{-1}$

2. Answer given to 3 significant figures with correct units

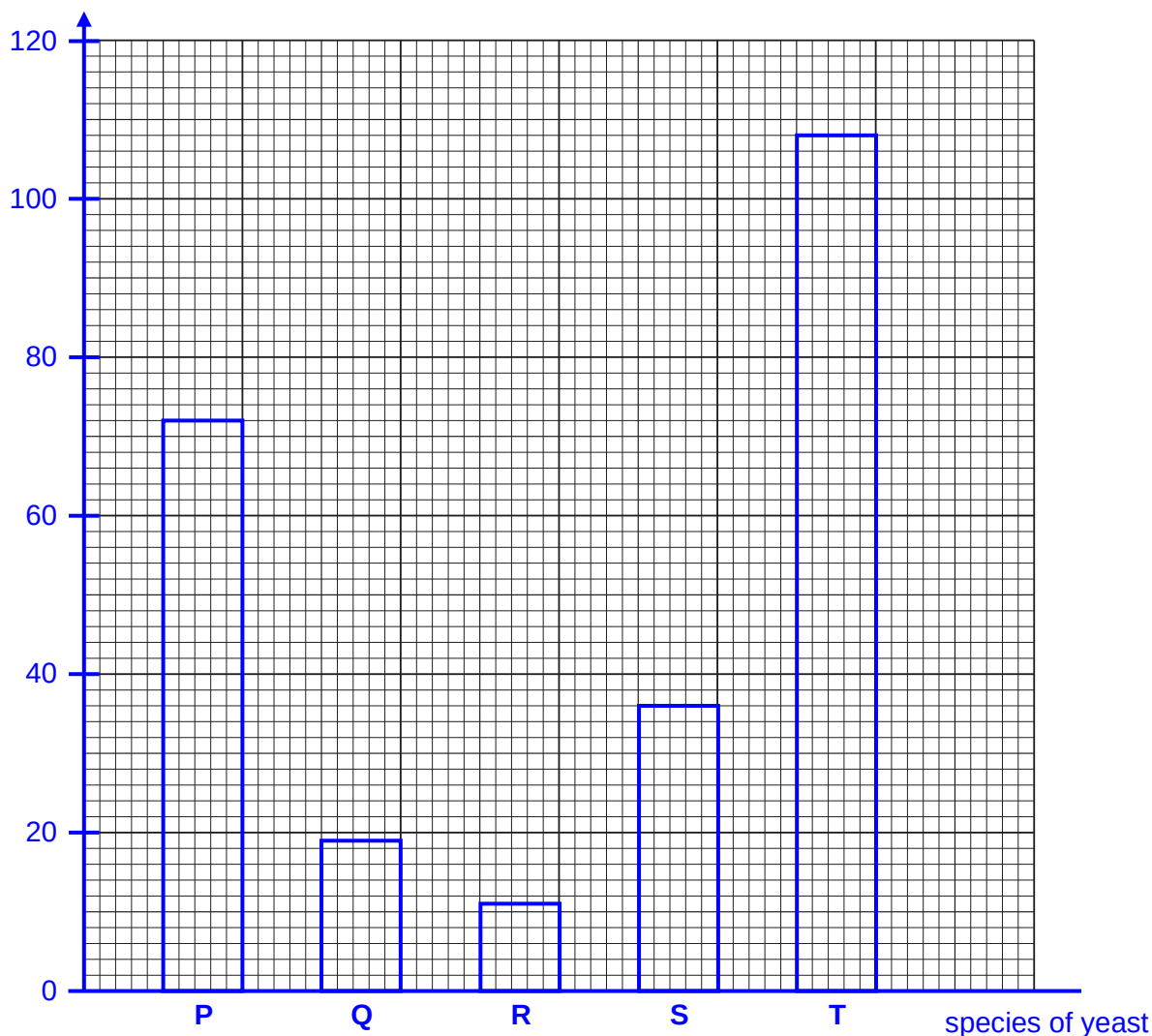
rate = ..... **0.0909 s<sup>-1</sup> / 1.82 cm s<sup>-1</sup>** ..... [2]

(ii) Use the grid to display the results shown in Table 1.3 in an appropriate form.

[4]

**Bar chart showing mean time to rise to surface for different species of yeast**

mean time to rise / s



- Scale** – bars of equal widths and spaced at equal intervals apart;  
every 5-10 boxes on y-axis labelled with numerical values starting with 0;  
R: no origin labelled, no 120 labelled
- Axes** – axes lines drawn with proper titles and correct units:  
y-axis: mean time to rise / s      x-axis: species of yeast;  
R: axis line not drawn (e.g. horizontal line for x-axis)
- Plot** – correct heights of each bar according to values in Table 1.3;
- Line** – clear, thin, sharp lines + labelling of bars P, Q, R, S, T;

(iii) Suggest why the results differ between species R and T.

[3]

1+2+3 OR 1+4+5

- Species R had the faster mean time to rise at 11 s compared to species T at 108 s;
- Higher gene expression of respiratory enzymes in species R than species T;
- Therefore rate of CO<sub>2</sub> production by respiration is higher for R than T;
- OR
- Species R has respiratory enzymes that are more active than species T;

5. Because the experiment was conducted at a temperature closer to optimum temperature of species R than species T;

6. AVP;

Species R has more mitochondria than species T → higher rate of respiration (A: decarboxylation in link reaction & Krebs cycle), and hence, produces more CO<sub>2</sub>

(c) Methylene blue stains dead cells, but is decolourised by living cells so they will appear white or clear.

You are required to observe the colour of methylene blue when added to three different yeast suspensions, **A1**, **A2** and **A3**.

It is known that each yeast suspension may be one of the following:

- boiled yeast,
- yeast in a high concentration of sodium chloride solution,
- yeast in a glucose solution.

Using this information, it is possible to deduce the identity of each yeast suspension.

**Proceed as follows.**

- 1 Label three microscope slides **A1**, **A2** and **A3**.
- 2 Place **one drop** of **A1** onto slide **A1** and add **one drop** of methylene blue.
- 3 Use a glass rod to mix carefully.
- 4 Leave for five minutes.
- 5 Repeat steps 2 to 4 with samples **A2** and **A3**.
- 6 Add a coverslip to each slide.
- 7 Use the paper towel to dry off any liquid around the coverslip.
- 8 View the slides using the microscope.

(i) Deduce the identity of each yeast suspension by filling in the table below with **A1**, **A2** or **A3**.

| description                                               | yeast suspension |
|-----------------------------------------------------------|------------------|
| boiled yeast                                              | <b>A2</b>        |
| yeast in a high concentration of sodium chloride solution | <b>A3</b>        |
| yeast in a glucose solution                               | <b>A1</b>        |

1 or 2 correct – 1 mark

All 3 correct – 2 marks

[2]

(ii) Explain the **difference in appearance** of the yeast cells in **A1** and **A2**.

[2]

1. Cells in A2 (boiled yeast) are dead, and hence would be stained blue;
2. While most cells in A1 are alive, and hence will decolourise methylene blue to appear colourless;

R: yeast cells are dead/alive (without any link to appearance)

[Total: 29]



2 L1 is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

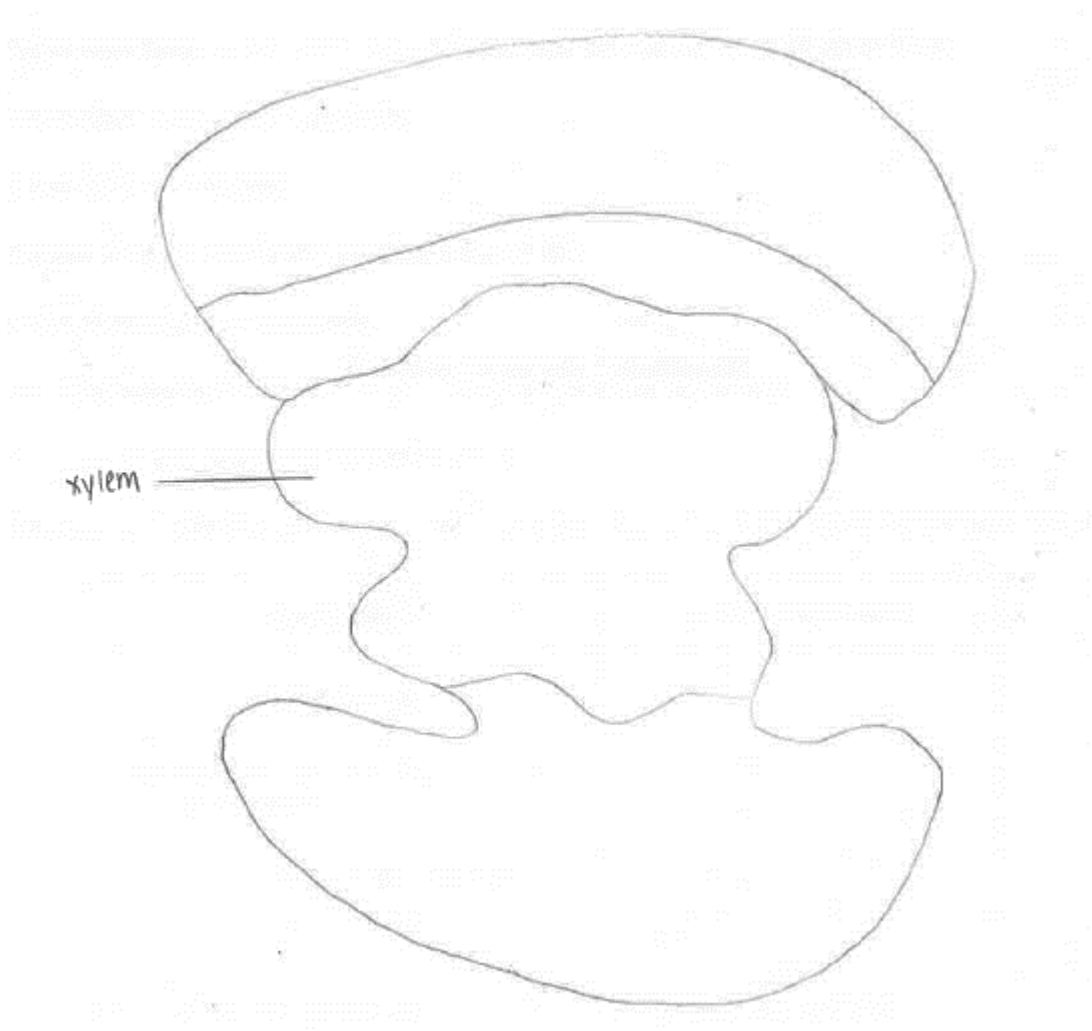
(a) Select a field of view so that you can observe a vascular bundle on L1.

Use a sharp pencil for drawing.

You are expected to draw the correct shape and proportions of the different tissues.

(i) Draw a large plan diagram of a vascular bundle on L1.

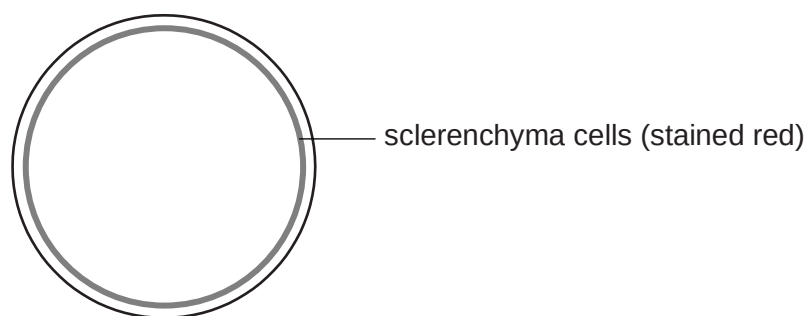
Use **one** ruled label line and label to identify the xylem.



[5]

|           |                                                                                     |
|-----------|-------------------------------------------------------------------------------------|
| <b>S</b>  | Smooth, thin, sharp, continuous lines + no shading<br>Size of drawing suitable      |
| <b>D1</b> | Draws vascular bundle only + no cells drawn                                         |
| <b>D2</b> | At least 4 layers/regions drawn as shown in mark scheme + general shape is accurate |
| <b>P</b>  | Proportion of each layer/region accurately represented                              |
| <b>L</b>  | Label line and label to xylem region correctly                                      |

Observe the sclerenchyma cells of the stem section on **L1**. These cells are stained red and are located towards the periphery of the stem section, as indicated in Fig. 2.1.

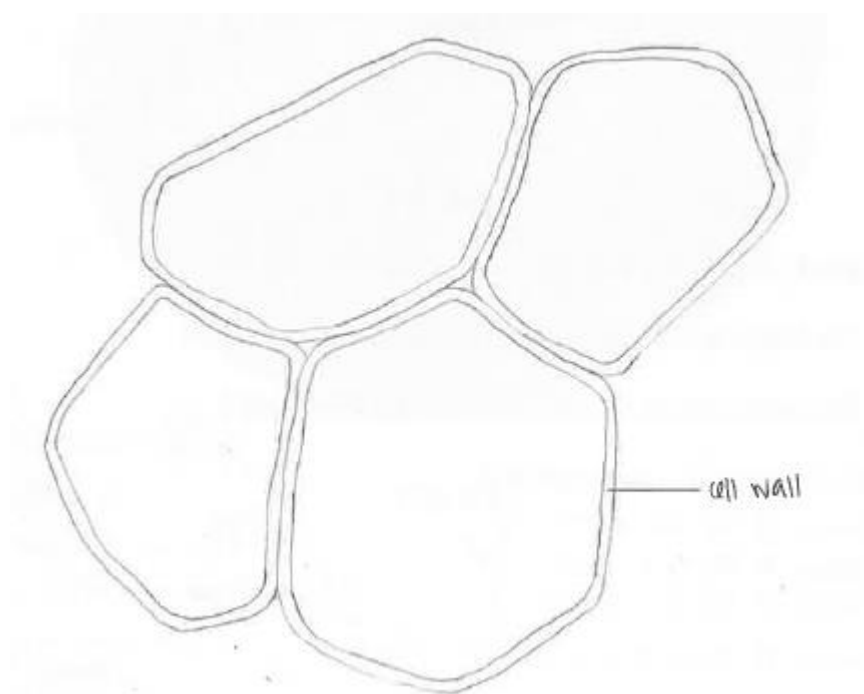


**Fig. 2.1**

Select a **group** of **four** adjacent sclerenchyma cells.

Each cell must touch at least two other cells.

- Make a large drawing of this group of **four** adjacent cells.
- Use **one** ruled label line and label to identify the cell wall of **one** cell.



[5]

|           |                                                                                                                                                                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>S</b>  | Smooth, thin, sharp, continuous lines + no shading<br>Size of drawing suitable                                                                                                                                                                            |
| <b>D1</b> | Draws only 4 sclerenchyma cells + each cell touching at least two other cells                                                                                                                                                                             |
| <b>D2</b> | Draws irregular/angular shape of (at least 2) cells – R: perfectly rectangular/round cells<br>Cell wall drawn as two lines around each cell + 3 lines where the cells touch<br>No big gap between cells<br>I: if inclusions drawn (e.g. nucleus, vacuole) |
| <b>P</b>  | Proportion of cells relatively equal + correct cell wall thickness – R: too thin or too thick                                                                                                                                                             |
| <b>L</b>  | Label line and label to <b>one</b> cell wall correctly                                                                                                                                                                                                    |

- (b) Fig. 2.2 is a photomicrograph of a stained transverse section through a stem of a different type of plant.

You are not expected to be familiar with this specimen.

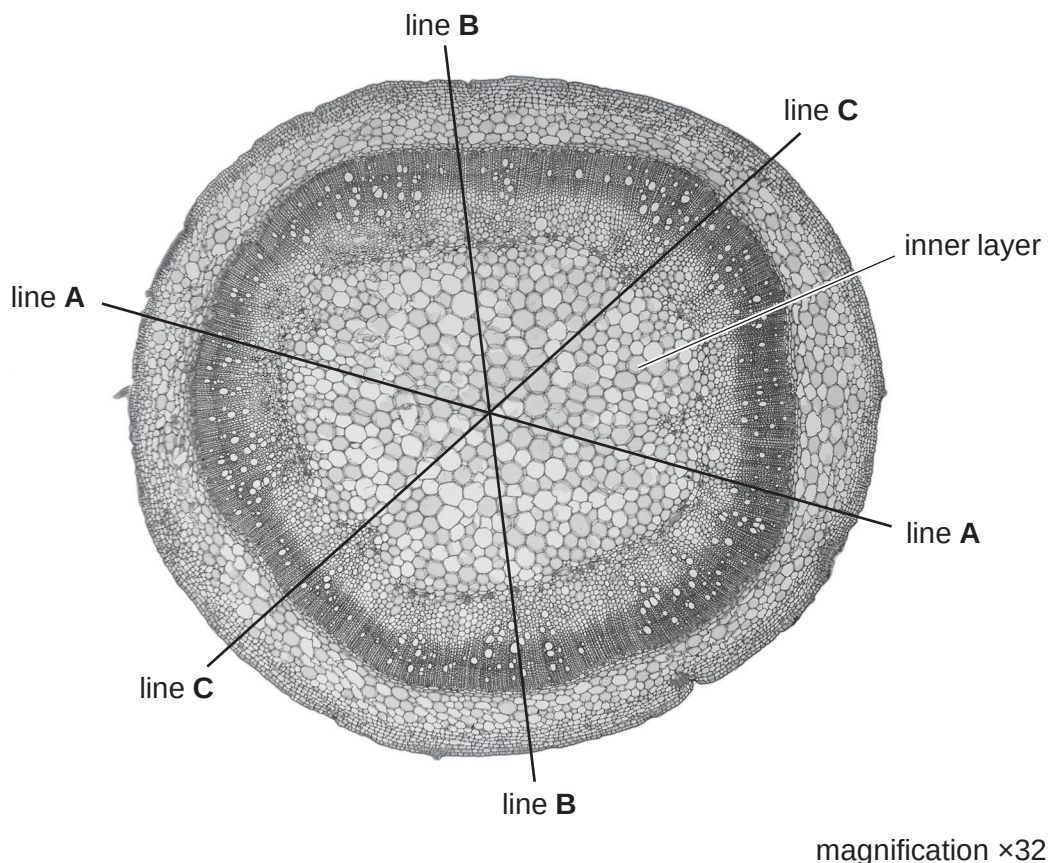


Fig. 2.2

- (i) Use line A, line B and line C to determine:
- the mean actual diameter of the whole stem section
  - the mean actual diameter of the inner layer (pith).

Show all the steps in your working.

[4]

Diameter of whole stem section = (98 mm + 90 mm + 97 mm) = 285 mm ;

A: 99 mm + 90 mm + 97 mm = 286 mm

Mean actual diameter of whole stem section =  $285000 \mu\text{m} / 3 / 32 = 2969 \mu\text{m}$  ;

A: 286000  $\mu\text{m}$  / 3 / 32 = 2979  $\mu\text{m}$

A: 58 mm + 45 mm + 47 mm = 150 mm

A: 58 mm + 46 mm + 47 mm = 151 mm

A: 59 mm + 45 mm + 47 mm = 151 mm

A: 59 mm + 46 mm + 47 mm = 152 mm

A: 60 mm + 45 mm + 47 mm = 152 mm

Diameter of inner layer = (60 mm + 46 mm + 47 mm) = 153 mm ;

A: 58 mm + 45 mm + 47 mm = 150 mm

A: 58 mm + 46 mm + 47 mm = 151 mm

A: 59 mm + 45 mm + 47 mm = 151 mm

A: 59 mm + 46 mm + 47 mm = 152 mm

A: 60 mm + 45 mm + 47 mm = 152 mm

Mean actual diameter of inner layer =  $153000 \mu\text{m} / 3 / 32 = 1594 \mu\text{m}$ ; A: 1590

A:  $150000 \mu\text{m} / 3 / 32 = 1563 \mu\text{m}$

A:  $151000 \mu\text{m} / 3 / 32 = 1573 \mu\text{m}$

A:  $152000 \mu\text{m} / 3 / 32 = 1583 \mu\text{m}$

**Marking points:**

1. Total diameter of whole stem;
2. Total diameter of inner layer;
3. Total diameter of whole stem and inner layer, both divide by 3 and 32;
4. Final answer to correct degree of accuracy (A: to 3 s.f) + units as  $\mu\text{m}$

mean actual diameter of whole stem section ..... **2969  $\mu\text{m}$** .....

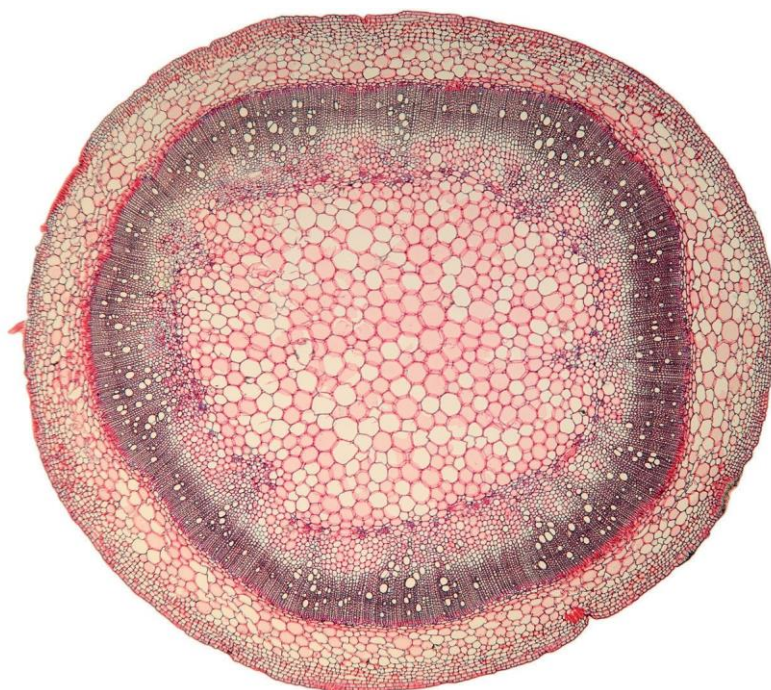
mean actual diameter of inner layer ..... **1594  $\mu\text{m}$** .....

- (ii) Use your answers to (b)(i) to determine the **simplest whole number ratio** of the mean actual diameter of the whole stem section to the mean actual diameter of the inner layer.

simplest whole number ratio = ..... **2:1**..... [1]

[ECF: single digit ratio based on answer in (b)(i)]

- (c) Fig. 2.3 is a photomicrograph of the same stem section that is in Fig. 2.2.



**Fig. 2.3**

Identify the observable differences between the stem section in Fig. 2.3 and the stem section on L1.

Record **two** observable differences in Table 2.1.

[2]

Table 2.1

| feature                                                       | Fig. 2.3                   | L1                                     |
|---------------------------------------------------------------|----------------------------|----------------------------------------|
| arrangement of vascular bundles                               | continuous                 | distinct<br>A: discontinuous           |
| shape of stem                                                 | circular                   | irregular<br>R: flower, star, pentagon |
| cavity in pith/inner region<br>A: presence or absence of pith | absent<br>(no hollow pith) | present<br>(hollow pith)               |

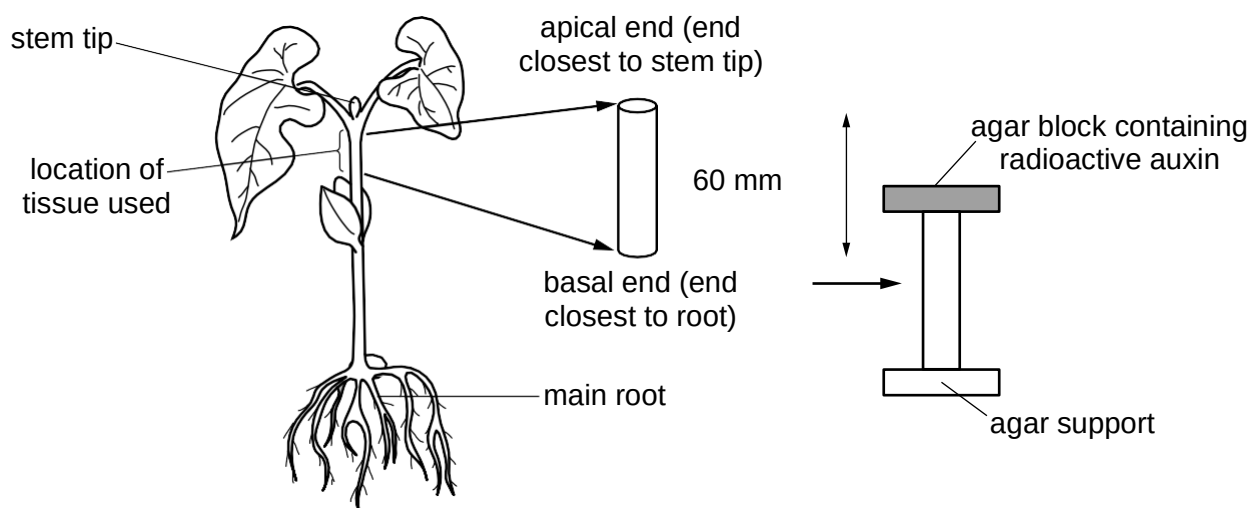
R:

- comparison of number of vascular bundles
- comparison of epidermis thickness

- (d) In plants the growth regulator, auxin, is synthesised in the stem tip and moves away from the tip. The movement of auxin through plant tissues was investigated using bean seedlings as shown in Fig. 2.4.

The following procedure was used.

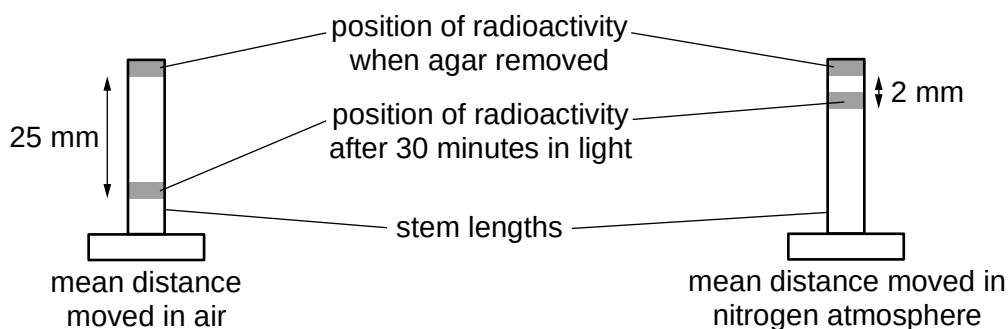
- Stems were cut into 60 mm lengths.
- Agar blocks containing radioactive auxin were placed on the apical surfaces of two groups of stem lengths.
- The basal ends of the stem lengths were placed on agar blocks without any auxin to provide support.



**Fig. 2.4**

- After 10 minutes the agar blocks at the apical ends were removed.
- One group of stem lengths was placed in air and the other group in an atmosphere of nitrogen.
- Both groups were left in light for 30 minutes after removing the agar blocks.
- The position of the radioactivity was located.

Fig. 2.5 shows the results of the investigation.



**Fig. 2.5**

- (i) Identify one variable that should be standardised during this investigation. [1]

Temperature; light intensity; volume/concentration of auxin; composition/type of agar; concentration of agar;

- (ii) Suggest **one** conclusion that can be made from these results. [1]

oxygen / air is required for the movement of auxin ;

A:

- rate of movement of auxin is faster in air than nitrogen
- distance moved by auxin is faster in air than nitrogen

energy / ATP / respiration is required for the movement of auxin ;

movement of auxin involves active transport ;

- (iii) Calculate the rate in  $\text{mm h}^{-1}$  of movement of auxin for the setup placed in air. [1]

$$25\text{mm}/0.5\text{h} = 50\text{mmh}^{-1}$$

A similar investigation was carried out to test the hypothesis:

The rate of movement of auxin will be faster in plants grown in the light than plants grown in the dark.

Table 2.2 shows the results of this investigation.

**Table 2.2**

| plants grown in light                              |    |    |    |    |    |    |    |    |    | plants grown in the dark                           |    |    |    |    |    |    |    |    |    |
|----------------------------------------------------|----|----|----|----|----|----|----|----|----|----------------------------------------------------|----|----|----|----|----|----|----|----|----|
| sample number                                      |    |    |    |    |    |    |    |    |    | sample number                                      |    |    |    |    |    |    |    |    |    |
| 1                                                  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 1                                                  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| rate of movement / $\text{mm h}^{-1}$              |    |    |    |    |    |    |    |    |    |                                                    |    |    |    |    |    |    |    |    |    |
| 56                                                 | 61 | 66 | 52 | 50 | 68 | 76 | 51 | 55 | 64 | 45                                                 | 52 | 42 | 35 | 55 | 38 | 32 | 37 | 45 | 51 |
| mean $\pm$ standard deviation (s) = $59.9 \pm 8.5$ |    |    |    |    |    |    |    |    |    | mean $\pm$ standard deviation (s) = $43.2 \pm 7.7$ |    |    |    |    |    |    |    |    |    |

- (iv) Describe how the following evidence from Table 2.2 supports the hypothesis. [2]

mean mean rate of movement in dark is slower than that in light;

standard deviation No overlap in terms of using the range of values within one standard deviation away from the mean

A: the lowest rate of movement of auxin for plants grown in light is  $51.4 \text{ mmh}^{-1}$ , which is still higher than the highest rate of auxin movement in plants grown in the dark of  $50.9 \text{ mmh}^{-1}$ .

- (v) A *t*-test was carried out to see if the difference in the rates of movement of auxin in plants grown in the light and plants grown in the dark was significant.

Suggest a null hypothesis for this statistical test. [1]

There is no significant difference in the rate of movement (of auxin) for, plants grown in light and (plants grown) in dark /between the 2 treatments;



(vi) Explain how the student should use the value for  $t$  to find out if the difference in the rates of movement of auxin is significant. [3]

1. finding / using degrees of freedom = 18;
2. from table of probabilities, determine p-value based on degrees of freedom and calculated t-value;  
A: compare the calculated t-value with the critical value at 5% significance level;
3. Reject null hypothesis if p is less than 0.05 at 5% significance level;  
A: if value of t is higher than critical value, reject null hypothesis;

[Total: 26]